Analysis of Crossover Designs with repeated measurements using GEE

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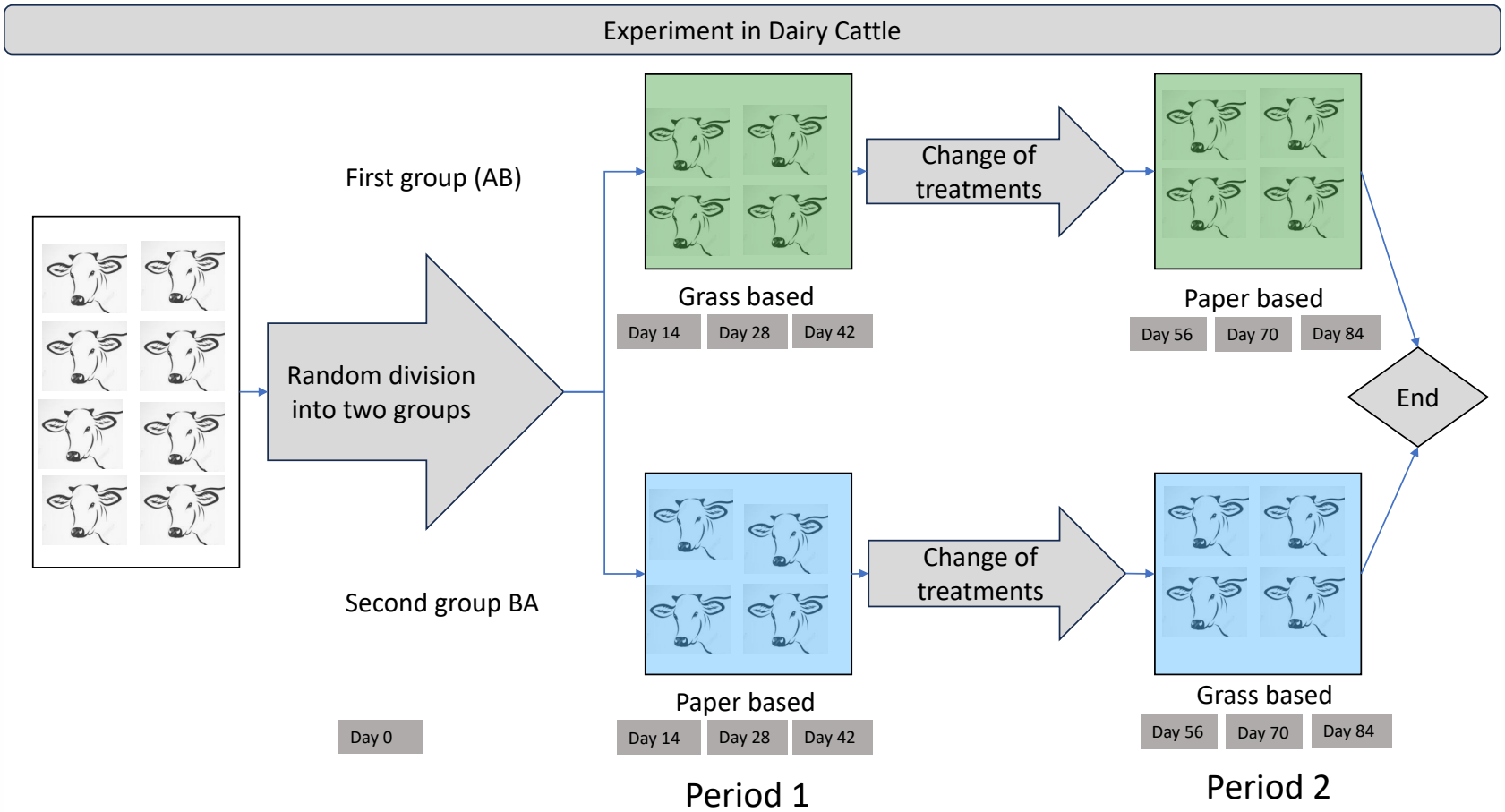
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Motivation


Experiment in Dairy Cattle
First group (AB)
Random division into two groups
Second group BA
Day 0
Grass based
Day 14
Day 28
Day 42
Change of treatments
Paper based
Day 56
Day 70
Day 84
End
Paper based
Day 14
Day 28
Day 42
Change of treatments
Grass based
Day 56
Day 70
Day 84
Period 1
Period 2

Crossover design in cows

An experiment carried out by researchers of the Department of Animal Sciences of Universidad Nacional de Colombia was focused on inferring the effects of two diets, A (grass) and B (a mixture of grass and paper recycling waste) on dairy cattle performance. Measurements of milk protein yield were taken (Jaime, 2019).

Sequence		Period 1	Period 2
(1) AB	Ind 1	3 measurements	3 measurements
	⋮	⋮	⋮
	Ind 4	3 measurements	3 measurements
(2) BA	Ind 5	3 measurements	3 measurements
	⋮	⋮	⋮
	Ind 8	3 measurements	3 measurements

Table: *Structure of the crossover design in cows*

Crossover design in arterial pressure

(Jones and Kenward, 2015, pg 204) described a crossover experiment with three treatments to control arterial pressure: treatment A is a placebo, and treatments B and C are 20 and 40 mg doses of a test drug. Each one of them was applied to two individuals lasting six weeks each and the measurement was taken in the middle of each period. Due to ethical reasons, there was no washout period between the treatments.

Sequence		Period 1	Period 2	Period 3
(1) ABC	Ind 1	10	10	10
	Ind 2	10	10	10
⋮	⋮	⋮	⋮	⋮
(6) CBA	Ind 11	10	10	10
	Ind 12	10	10	10

Table: *Structure of the blood pressure crossover design*

10 consecutive measurements of diastolic arterial pressure were taken: -30, -15, 15, 30, 45, 60, 75, 90, 120 and 240 minutes after the administration of the treatment

Arterial data

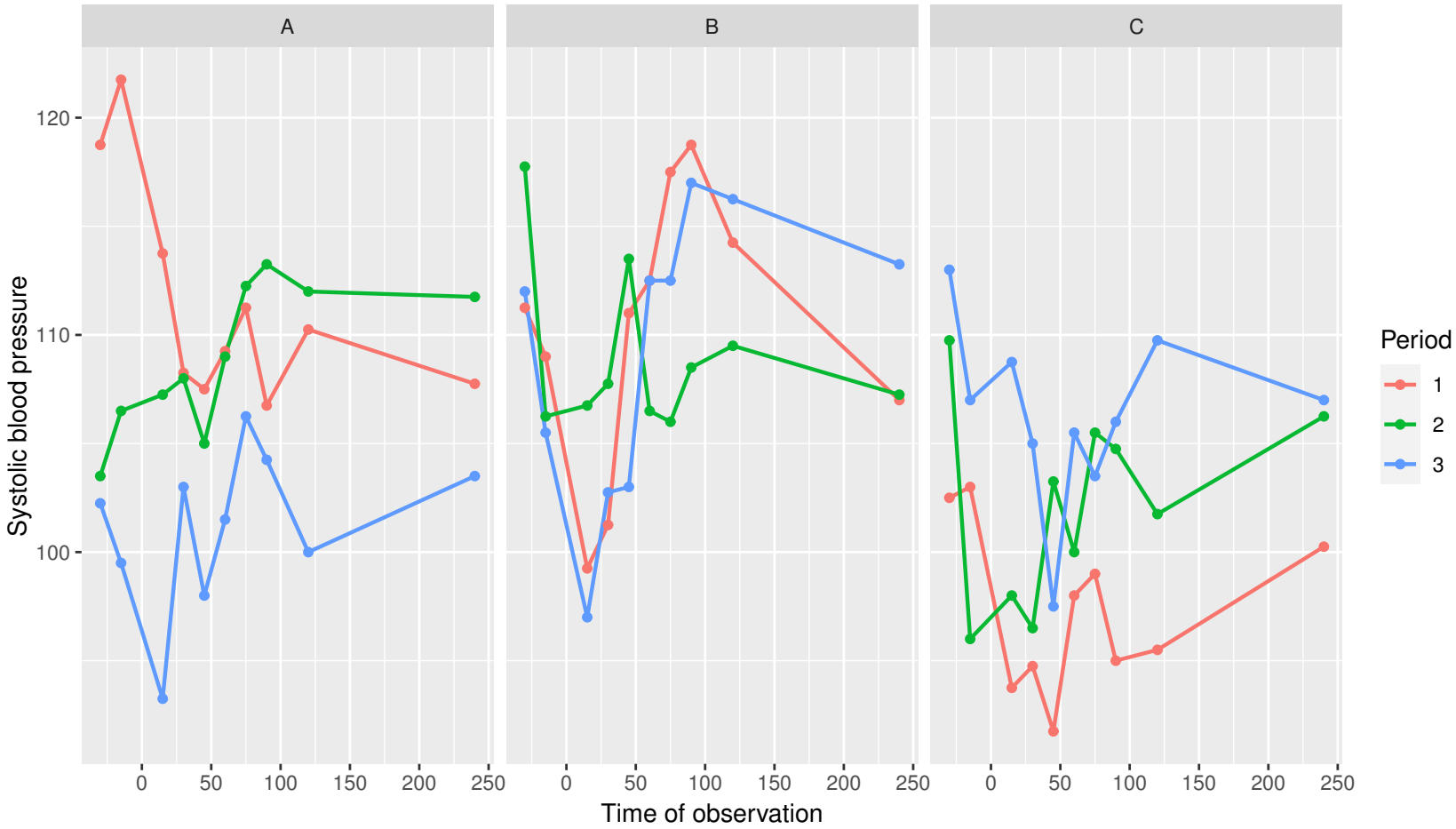


Figure: *Mean arterial pressure of the patients in the blood pressure experiment*

Crossover design

A crossover design has the following components (Patterson, 1951):

Components

1. *Sequences*: which are each one of the distinct treatment combinations to be sequentially and randomly applied to the experimental units.
2. *Treatments*: randomly applied to each experimental unit within each sequence.
3. *Periods*: the time intervals in which the treatments that make up a sequence are applied, usually, each period is the same for all sequences and, consequently, the number of periods is equal to the number of elements of each sequence.
4. *Experimental unit*: the subjects or elements that receive the treatments, the total number of experimental units in the study is denoted by $\boldsymbol{n}$.

Carryover effects

Carryover effects

Persistency of the effect of a treatment over those that will be applied later, i.e., the treatment is applied in a given period, but its effect still affects the response in later periods when other treatments are applied. When the effect persists after one period, it is known as first order carryover effect, and when it persists after two periods, it is called a second order carry-over effect and so on (Vegas et al., 2016).

Types

1. **Simple carry-over**: where the residual effect of treatment affects equally each of the treatments that are preceded by it, and
2. **Complex carry-over**: where the residual effect of the treatment affects each of the other treatments differently.

Background

1. In Curtin (2017) and Li et al. (2018), generalized linear models were used for crossover designs for two periods and two sequences of two treatments where the experimental unit was observed once in each period.
2. Josephy et al. (2015), Hao et al. (2015), Lui (2015), Grayling et al. (2018), Madeyski and Kitchenham (2018) and Kitchenham et al. (2018) used mixed models for crossover designs with a normal response and a single observation per period, including additively carry-over effects in the model.
3. Diaz et al. (2013) and Forbes et al. (2015) normal mixed models are used to study crossover designs with repeated measures and without including carry-over effects.
4. Generalized estimating equations (GEE) were used by Jankar et al. (2020) to estimate model parameters in crossover experiments.

Outline

Problems

- There is no adequate methodology to perform a deviance analysis table with carryover effects.
- It is necessary to build an inference that performs well in scenarios of a few experimental units.
- The usual correlation structures are very restrictive if the number of observations per period is large.
- Assuming that the carry-over effects are constant over time is very naive.

Solutions

1. A Deviance Analysis methodology was constructed.
2. A new correlation structure using the Kronecker product was developed.
3. A semiparametric GEE model was developed to estimate the temporal and carry-over effects.

Notation

In a crossover design with S sequences of length (the number of elements comprising each sequence) P, Y_{ijk} is defined as the kth response of the ith experimental unit at jth period. Let n_{ij} be the number of observations of the ith experimental unit during the jth period. Then, vector $\boldsymbol{Y}_{ij}$ is defined as:

$$\boldsymbol{Y}_{ij} = \left(Y_{ij1}, \ldots, Y_{ijn_{ij}}\right)^T$$

Also, vector $\boldsymbol{Y}_i$ is defined as:

$$\boldsymbol{Y}_i = \left(\boldsymbol{Y}_{i1}, \ldots, \boldsymbol{Y}_{iP}\right)^T$$

which has dimension $\sum_{j=1}^{P} n_{ij}$, where P is the number of periods.

$$i = 1, \ldots, n$$
$$j = 1, \ldots, P$$
$$k = 1, \ldots, n_{ij}$$
$$L = \max_{ij}\{n_{ij}\}$$

Why using GEE?

1. Assuming that $\boldsymbol{Y_{ijk}}$ has a distribution in the exponential family, generalized linear mixed models (GLMM) or generalized estimating equations (GEE) can be used.
2. While the GLMM explicitly models within-subject correlation using random effects, the GEE implicitly explains such correlations using sandwich variance estimates (Liang and Zeger, 1986).
3. The GEE estimators yield a more solid inference about the fixed effects than the GLMM that depends on the distributional assumptions about the random effects. (Zhang et al., 2012).

GEE

Since the main goal of a crossover experiment is to obtain a robust inference about treatment effects, GEE models are more suitable than GLMM models.

Notation

Model

$$E(Y_{ijk}) = \mu_{ijk} \qquad g(\mu_{ijk}) = \boldsymbol{x}_{ijk}^T \boldsymbol{\beta} = \eta_{ijk}$$

$$\eta_{ijk} = \mu + \text{Period} + \text{Treatment} + \text{1st order carryover}$$

$$\eta_{ijk} = \mu + \gamma_j + \tau_{d[i,j]} + \theta_{d[i,j-1]} + \cdots + \theta_{d[i,j-P+1]}$$

where:

- $g(\cdot)$ is the link function.
- $\boldsymbol{x}_{ijk}$ is the vector of the design matrix corresponding to kth response from the ith experimental unit at the jth period.
- $\boldsymbol{\beta}$ is the vectors of fixed effects.
- $\tau_{d[i,j]}$ is the effect of treatment and $\theta_{d[i,j-u]}$ is the carry-over effect of order u

GEE

Because observations of the same experimental unit are correlated, parameter estimation is carried out using GEE (Liang and Zeger, 1986).

Estimating equation for $\boldsymbol{\beta}$

$$\boldsymbol{U(\beta)} = \left[\left\{ \sum_{i=1}^{n_i} \boldsymbol{x}_{mi}^T \boldsymbol{D} \left(\frac{\partial \boldsymbol{\mu}_i}{\partial \eta} \right) [\boldsymbol{V}(\boldsymbol{\mu}_i)]^{-1} \left(\frac{\boldsymbol{y}_i - \boldsymbol{\mu}_i}{\phi} \right) \right\}_{m=1,\ldots,q} \right]_{q \times 1}$$

where:

- $\boldsymbol{\mu}_i = (\mu_{i11}, \ldots, \mu_{iPn_{iP}})^T$.
- $\boldsymbol{y}_i = (y_{i11}, \ldots, y_{iPn_{iP}})^T$.
- $\boldsymbol{x}_{mi}$ is the $\boldsymbol{m}$th column of the design matrix of the ith experimental unit.
- ϕ is the dispersion parameter.
- $\boldsymbol{n}_i = \sum_{j=1}^{P} \boldsymbol{n}_{ij}$ is the total number of observations of the $\boldsymbol{i}$th experimental unit.

Variance component

$V(\mu_i)$

$$V(\mu_i) = \left[D(V(\mu_{ijk})^{\frac{1}{2}}) R(\alpha) D(V(\mu_{ijk})^{\frac{1}{2}}) \right]_{P \times P}$$

in the case that Y_{ijk} is binary, $V(\mu_{ijk}) = \mu_{ijk}(1 - \mu_{ijk})$ and $D(V(\mu_{ijk}))$ would be of the form:

$$D(V(\mu_{ijk})) = \begin{pmatrix} \mu_{i11}(1-\mu_{i11}) & 0 & \cdots & 0 \\ 0 & \mu_{i12}(1-\mu_{i12}) & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mu_{iPL}(1-\mu_{iPL}) \end{pmatrix}_{PL \times PL}$$

Correlation structure

$\boldsymbol{R(\alpha)}$ is the correlation matrix related to the covariance matrix $\boldsymbol{\Sigma_i = Var(Y_i)}$ as follows:

Correlation

$$(\boldsymbol{\Sigma}_i)_{PL\times PL} = \left(\boldsymbol{D}(Var(Y_{ijk})^{\frac{1}{2}})\boldsymbol{R(\alpha)}\boldsymbol{D}(Var(Y_{ijk})^{\frac{1}{2}})\right)_{PL\times PL}.$$

It is assumed that the correlation has a structure to more parsimoniously estimate $\boldsymbol{\alpha}$. For example:

$$\boldsymbol{R(\alpha)} = \begin{pmatrix} 1 & \alpha & \cdots & \alpha \\ \alpha & 1 & \cdots & \alpha \\ \vdots & \vdots & \ddots & \vdots \\ \alpha & \alpha & \cdots & 1 \end{pmatrix}_{PL\times PL} \qquad \boldsymbol{R(\alpha)} = \begin{pmatrix} 1 & \alpha & \cdots & \alpha^{PL-1} \\ \alpha & 1 & \cdots & \alpha^{PL-2} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha^{PL-1} & \alpha^{PL-2} & \cdots & 1 \end{pmatrix}_{PL\times PL}$$

Usual GEE

Estimating equation for $\boldsymbol{\beta}$

$$\boldsymbol{U}_1(\boldsymbol{\beta}) = \left[\left\{ \sum_{i=1}^{n_i} \boldsymbol{x}_{mi}^T \boldsymbol{D} \left(\frac{\partial \boldsymbol{\mu}_i}{\partial \eta} \right) [\boldsymbol{V}(\boldsymbol{\mu}_i)]^{-1} \left(\frac{\boldsymbol{y}_i - \boldsymbol{\mu}_i}{\phi} \right) \right\}_{m=1,\ldots,q} \right]_{q \times 1}$$

Estimating equation for $\boldsymbol{\alpha}$

$$\boldsymbol{U}_2(\boldsymbol{\alpha}) = \sum_{i=1}^{S} \sum_{k=1}^{n_i} \left(\frac{\partial \boldsymbol{\epsilon}_{ik}}{\partial \boldsymbol{\alpha}} \right)^t \boldsymbol{H}_{ik}^{-1} (\boldsymbol{W}_{ik} - \boldsymbol{\epsilon}_{ik})$$

where $\boldsymbol{H}_{ik} = \boldsymbol{D}(\boldsymbol{V}(r_{ijk}))_{q \times q}$ is a diagonal matrix, $\boldsymbol{\epsilon}_{ik} = \boldsymbol{E}(\boldsymbol{W}_{ik})_{q \times 1}$ and $\boldsymbol{W}_{ik} = (r_{i1k} r_{i2k}, r_{i1k} r_{i3k}, \ldots, r_{i(T-1)k} r_{iTk})^t_{q \times 1}$, with r_{ijk} is the ijk-th Pearson residual, and $q = \binom{T}{2}$.

Model selection

The ***QIC*** is defined as:

QIC

$$QIC = -2QL(\hat{\mu}; I) + 2trace(\hat{\Omega}_I^{-1}\hat{V}_R)$$

where:

- $\hat{\mu}_{ijk} = \hat{\eta}_{ijk} = g^{-1}(x_{ijk}\hat{\beta})$ is the estimated expected value of observation Y_{ijk} under the model assuming the correlation matrix R.
- $\hat{\Omega}_I$ is the estimated covariance matrix of β under independence.
- $\hat{V}_R$ is the covariance matrix of β under the model with correlation matrix $R(\alpha)$.

Use

After fitting several models with the same formula, but with different correlation structures, the model with the lowest ***QIC*** is selected because it presents the best balance between goodness-of-fit and complexity.

First Result

Theorem 1

Let $\hat{\boldsymbol{\beta}}$ be the solution of the GEE and $\boldsymbol{\beta} = (\boldsymbol{\beta}_1^t, \boldsymbol{\beta}_2^t)^t$. Under the null hypothesis $\boldsymbol{H_0} : \boldsymbol{\beta}_1 = \boldsymbol{\beta}_{10}$, the test statistic, $\boldsymbol{T_w}$, defined as:

$$\boldsymbol{T_w} = \boldsymbol{N}(\hat{\boldsymbol{\beta}}_1 - \boldsymbol{\beta}_{10})^t \boldsymbol{V}_{\hat{\boldsymbol{\beta}}_1}^{-1} (\hat{\boldsymbol{\beta}}_1 - \boldsymbol{\beta}_{10})$$

under $\boldsymbol{H_0}$ and under some regularity conditions, follows an asymptotic distribution $\boldsymbol{T_w} \sim \boldsymbol{F_{T_W}}$ with $\boldsymbol{F_{T_W}}$ defined as:

$$\boldsymbol{P}(\boldsymbol{T_W} \leq \boldsymbol{w}) = \boldsymbol{F_{T_W}}(\boldsymbol{w}) = \left(\prod_{i=1}^{p} \boldsymbol{b_i} \right) \sum_{j=0}^{\infty} \left(\boldsymbol{a_j} \int_0^{\boldsymbol{w}} \frac{\boldsymbol{y}^{\frac{1}{2}+j-1} \boldsymbol{e}^{-\frac{y}{2c_1}}}{(2\boldsymbol{c_1})^{\frac{1}{2}+j} \boldsymbol{\Gamma}(\frac{1}{2}+j)} \boldsymbol{dy} \right)$$

Deviance analysis

where c_i as the i-th eigenvalue of the matrix $\boldsymbol{Q} = \boldsymbol{Q}_0^{-1}\boldsymbol{Q}_1$. $\boldsymbol{Q}_0$ and $\boldsymbol{Q}_1$ are, respectively,

$$\boldsymbol{Q}_0 = \frac{1}{n}\sum_{i=1}^{S}\sum_{k=1}^{n_i}\tilde{\boldsymbol{D}}_{ik}\boldsymbol{V}(\boldsymbol{\mu}_i)^{-1}\boldsymbol{D}_{ik}$$

$$\boldsymbol{Q}_1 = \frac{1}{n}\sum_{i=1}^{S}\sum_{k=1}^{n_i}\tilde{\boldsymbol{D}}_{ik}\boldsymbol{V}(\boldsymbol{\mu}_i)^{-1}Cov(\boldsymbol{Y}_{ik})\boldsymbol{V}(\boldsymbol{\mu}_i)^{-1}\boldsymbol{D}_{ik}$$

$$\boldsymbol{Cov}(\boldsymbol{Y}_{ik}) = \boldsymbol{D}(\boldsymbol{\mu}_i)^{\frac{1}{2}}\left(\frac{1}{n}\sum_{i=1}^{S}\sum_{i=1}^{n_i}\boldsymbol{D}(\boldsymbol{\mu}_i)^{-\frac{1}{2}}\boldsymbol{S}_{ik}\boldsymbol{S}_{ik}^{t}\boldsymbol{D}(\boldsymbol{\mu}_i)^{-\frac{1}{2}}\right)\boldsymbol{D}(\boldsymbol{\mu}_i)^{\frac{1}{2}}$$

$$b_i = \sqrt{\frac{c_1}{c_i}}$$

$$A(c_i, r) = \left[\prod_{k=0}^{r-1}\left(\frac{1}{2}+k\right)\left(1-\frac{c_1}{c_r}\right)^{r}\right](r!)^{-1}$$

Sketch of proof

1. By Theorem 2 of Liang and Zeger (1986) it is known that $\hat{\boldsymbol{\beta}}$ converges in probability to $\boldsymbol{\beta}$, furthermore:

$$\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \xrightarrow{D} \boldsymbol{N}(\mathbf{0}, \boldsymbol{Var}(\hat{\boldsymbol{\beta}}))$$

2. A robust and consistent estimator for the variance of $\hat{\boldsymbol{\beta}}$ (Pan, 2001) is given by:

$$\begin{aligned}\widehat{\boldsymbol{Var}}(\hat{\boldsymbol{\beta}}) &= \left(\sum_{i=1}^{n} \boldsymbol{D}_i^t \boldsymbol{V}(\boldsymbol{\mu}_i)^{-1} \boldsymbol{D}_i\right)^{-1} \left(\sum_{k=1}^{n} \boldsymbol{D}_i^t \boldsymbol{V}(\boldsymbol{\mu}_i)^{-1} \widehat{\boldsymbol{Cov}}(\boldsymbol{y}_i) \boldsymbol{V}(\boldsymbol{\mu}_i)^{-1} \boldsymbol{D}_i\right) \\ &\quad \left(\sum_{k=1}^{n} \boldsymbol{D}_i^t \boldsymbol{V}(\boldsymbol{\mu}_i)^{-1} \boldsymbol{D}_i\right) \\ &= \boldsymbol{A}^{-1} \boldsymbol{B} \boldsymbol{A}\end{aligned}$$

Sketch of proof

3. For the Result 1 of Rotnitzky and Jewell (1990), it is obtained that the T_w statistic:

$$T_w = n(\hat{\beta}_1 - \beta_{10})^t V_{\hat{\beta}_1}^{-1}(\hat{\beta}_1 - \beta_{10}) \overset{n\to\infty}{\sim} \sum_{j=1}^{p} c_j \chi_j^2$$

4. Then, the distribution of $W = \sum_{j=1}^{p} c_j \chi_j^2$ is found based on Moschopoulos and Canada (1984) as follows:

$$F_W(w) = P(W \leq w) = \left(\prod_{i=1}^{p} b_i\right) \sum_{j=0}^{\infty} \left(a_j \int_0^w g_j(y)dy\right)$$

$$= \left(\prod_{i=1}^{p} b_i\right) \sum_{j=0}^{\infty} \left(a_j \int_0^w \frac{y^{\frac{1}{2}+j-1} e^{-\frac{y}{2c_1}}}{(2c_1)^{\frac{1}{2}+j}\Gamma(\frac{1}{2}+j)} dy\right) \square$$

Simulation study

Sequence	First Period	Second Period	Third Period
1	A	B	A
2	B	A	B

Table: *Switch back crossover design*

Simulation

- $Y_{ijk} \sim Poisson(\mu_{ijk})$; $i = 1, \ldots, n$; $n \in \{5, 10, 20, 50\}$
- $ln(\mu_{ijk}) = \mu + \tau +$ Period effect $+$ Carryover effect
- $\tau = \tau_B - \tau_A$, and $\tau \in \{-2, -1.9, -1.8, \ldots, 1.8, 1.9, 2\}$
- Period effect $\sim U(-1, 1)$ and carry-over effect $\sim U(-1, 1)$.
- The percentage of times the null hypothesis $H_0 : \tau = 0$ was rejected was calculated with two test statistics:
  1. The usual normal asymptotic
  2. The test statistic T_w proposed in this work

Simulation study

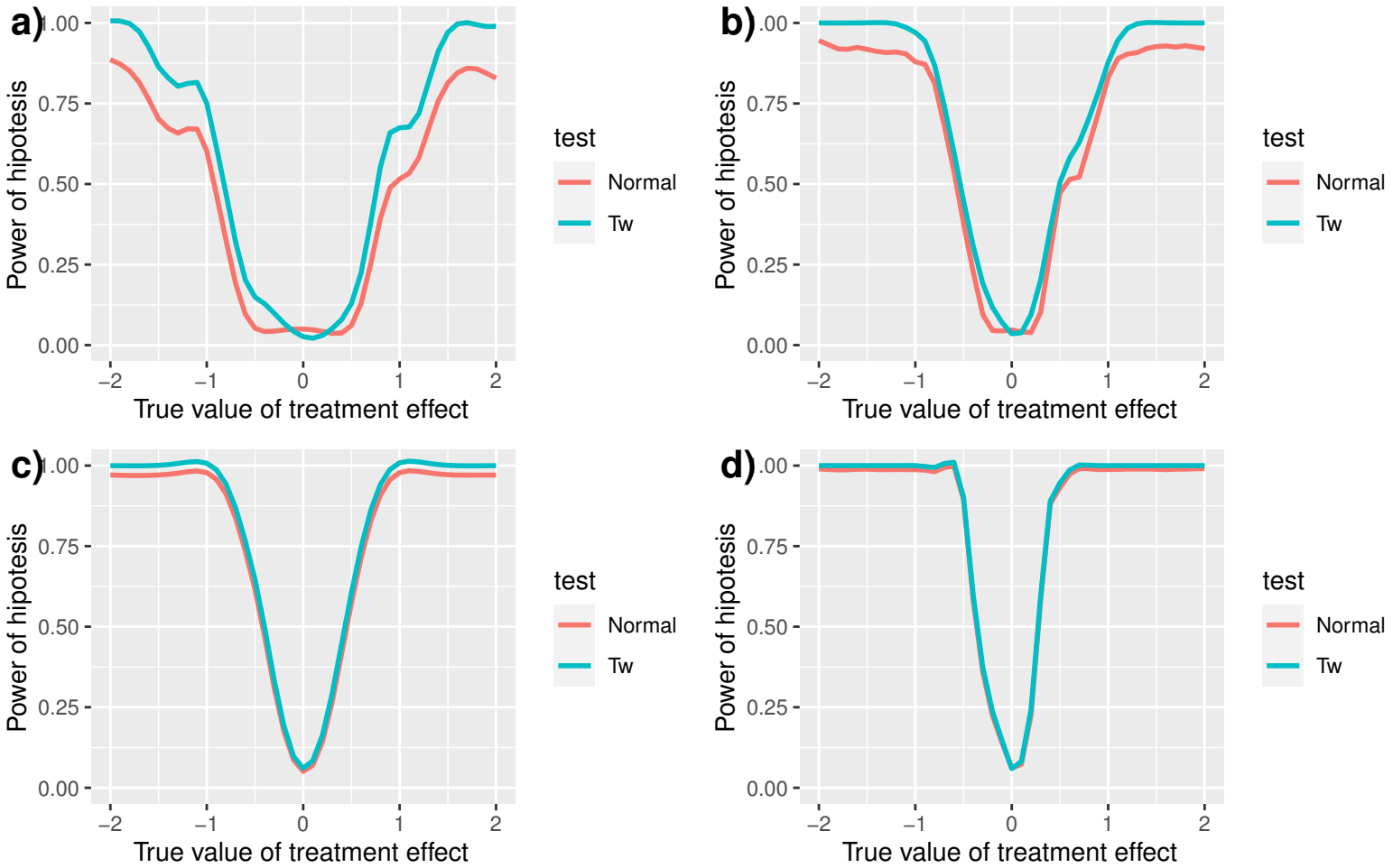


Figure: *Power of the test statistic for different values of the treatment effect* τ *under* H_0*;* $\tau = 0$*. The number of replications was: a)* **5***, b)* **10***, c)* **20***, and d)* **50***. In each subplot, the power was calculated in two ways, i)* T_w*: with the test statistic proposed in this work and ii)* ***Normal****: with the asymptotic usual result.*

Applications

Matrix $R(\alpha)$	QIC arterial	QIC cows
I_{PL}	46086	20.21
$AR(1)_{PL}$	44923	20.46
$Exch_{PL}$	44158	20.11
R_{PL}	100107	25.22

Table: *Correlation matrices and the corresponding* QIC *values*

Deviance table for Dairy cattle

Source	T_w	$P(> T_w)$
Period	32.686	$<$0.001
Treatment	16.839	$<$0.001
Carry_over	8.984	0.003
Time	2.049	0.152
Time^2	1.317	0.251

And for arterial data

Source	T_w	$P(> T_w)$
Period	3.140	0.118
Treatment	14.196	$<$0.001
Carry_over	3.450	0.154
Time	0.057	0.811
Time^2	0.437	0.508

Estimated values for each parameter of the model

Parameters in Dairy cattle

Parameter	Estimate	Std.err
(Intercept)	0.32507	0.09132
Periodo2	0.34236	0.05988
Treat_paper	-0.23583	0.05747
Carry_paper	-0.33917	0.11316
Time	-0.00819	0.00572
Time^2	0.00010	0.00009

$$\hat{\alpha} = \mathbf{0.563(0.239)}$$

$$\hat{\phi} = \mathbf{0.0101(0.00344)}$$

Parameters in arterial data

Parameter	Estimate	Std.err
(Intercept)	107.74144	3.19128
Period2	2.06035	1.31895
Period3	0.93535	1.12253
TreatmentB	1.55805	1.78015
TreatmentC	-6.26840	1.66384
Carry_B	-2.12586	1.78717
Carry_C	-3.10519	2.17855
Time	-0.00585	0.02441
Time^2	0.00006	0.00009

$$\hat{\alpha} = \mathbf{0.499(0.112)}$$

$$\hat{\phi} = \mathbf{136(23.1)}$$

Exchangeable correlation matrix

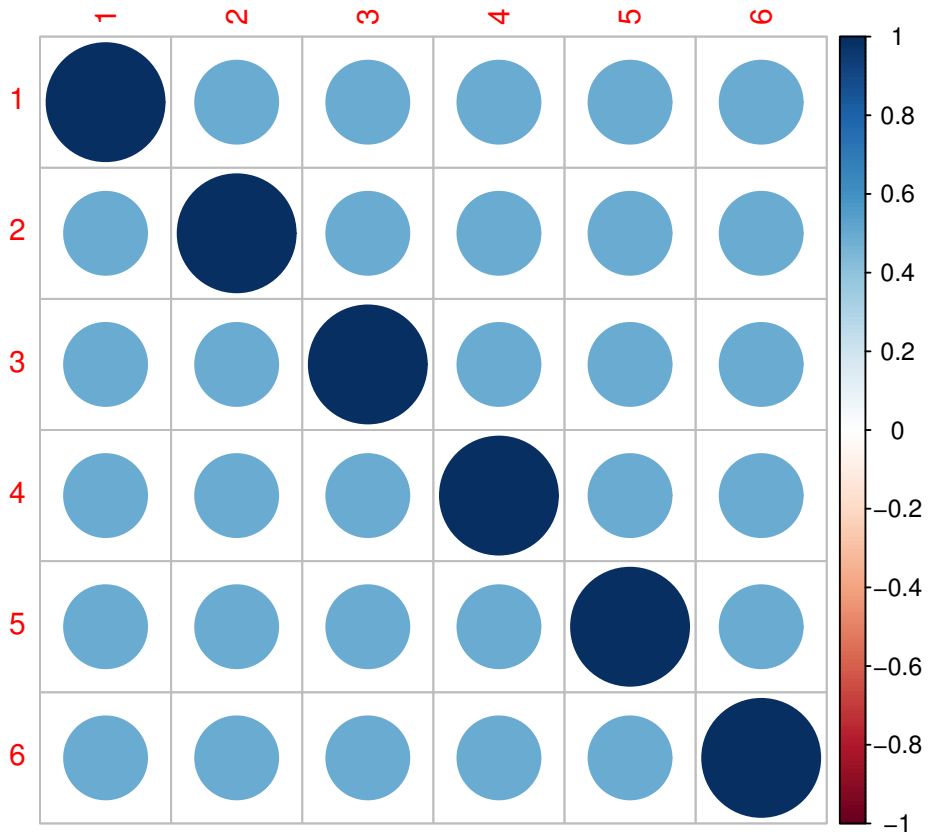


Figure: *Estimated correlation matrix for cow data.*

Exchangeable correlation matrix

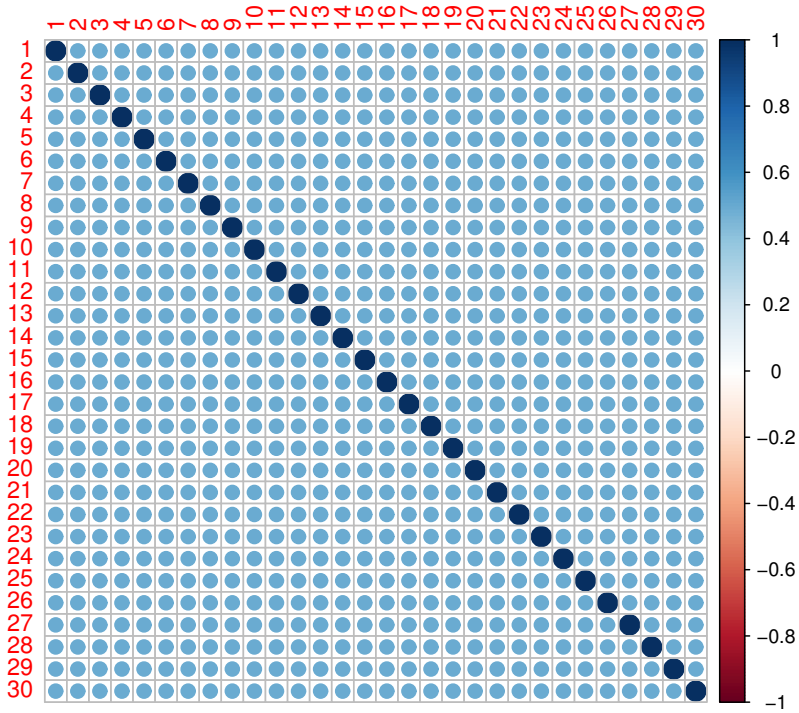


Figure: *Estimated correlation matrix for cow data.*

When we fit the correlation matrix for blood pressure, we have a 30 by 30 matrix. Either it has many parameters (Unstructured $= \frac{30 \times 29}{2} - \mathbf{30 = 405}$ parameters) or it is very simple (Autoregressive or Interchangeable $= 1$ parameter)

New correlation structure

We propose a correlation structure of the form:

Correlation

$$\boldsymbol{R}(\boldsymbol{\alpha}) = \boldsymbol{\Psi} \otimes \boldsymbol{R}_1(\boldsymbol{\alpha}_1)$$

where $\boldsymbol{R}_1(\boldsymbol{\alpha}_1)$ is the within-period correlation matrix, $\boldsymbol{\Psi}$ is the between-period correlation matrix, and $\otimes$ represents the Kronecker product (Harville, 1997). If $\boldsymbol{P} = 3$ y $\boldsymbol{L} = 10$, an example of $\boldsymbol{R}(\boldsymbol{\alpha})$ would be:

$$\boldsymbol{R}_1(\boldsymbol{\alpha}_1) = \begin{pmatrix} 1 & \alpha_1 & \alpha_1 & \dots & \alpha_1 \\ \alpha_1 & 1 & \alpha_1 & \dots & \alpha_1 \\ \alpha_1 & \alpha_1 & 1 & \dots & \alpha_1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \alpha_1 & \alpha_1 & \dots & \alpha_1 & 1 \end{pmatrix}_{10\times 10}, \quad \boldsymbol{\Psi} = \begin{pmatrix} 1 & \psi_{12} & \psi_{13} \\ \psi_{12} & 1 & \psi_{23} \\ \psi_{13} & \psi_{23} & 1 \end{pmatrix}_{3\times 3}$$

Kronecker GEE

Estimating equation for $\boldsymbol{\beta}$

$$\boldsymbol{U}_1(\boldsymbol{\beta}) = \left[\left\{ \sum_{i=1}^{n_i} \boldsymbol{x}_{mi}^T \boldsymbol{D} \left(\frac{\partial \boldsymbol{\mu}_i}{\partial \eta} \right) [\boldsymbol{V}(\boldsymbol{\mu}_i)]^{-1} \left(\frac{\boldsymbol{y}_i - \boldsymbol{\mu}_i}{\phi} \right) \right\}_{m=1,\ldots,q} \right]_{q \times 1}$$

Estimating equation for $\boldsymbol{\alpha}_1$

$$\boldsymbol{U}_2(\boldsymbol{\alpha}_1) = \sum_{i=1}^{n} \left(\frac{\partial \boldsymbol{\varepsilon}_i}{\partial \boldsymbol{\alpha}_1} \right)^T \boldsymbol{H}_i^{-1} (\boldsymbol{W}_i - \boldsymbol{\varepsilon}_i)$$

Estimating equation for $\boldsymbol{\Psi}$

$\boldsymbol{\Psi} = (\psi_{jj'})_{P \times P}$

$$\hat{\psi}_{jj'} = \frac{1}{n} \sum_{i=1}^{n} tr\left(\boldsymbol{R}_1(\hat{\boldsymbol{\alpha}}_1)^{-1} (\boldsymbol{r}_{(j)i} - \bar{\boldsymbol{r}}_{(j)})(\boldsymbol{r}_{(j')i} - \bar{\boldsymbol{r}}_{(j')})^T \right)$$

where:

- $\boldsymbol{r}_{(j)i}$ is the vector of Pearson residuals for the ith experimental unit at the jth period.
- $\bar{\boldsymbol{r}}_{(j)}$ is the average (over i), $j \neq j'$, with $j = 1, \ldots, P$.

Steps

1. $\boldsymbol{R}(\boldsymbol{\alpha}_1)$ is estimated with the residuals grouped by experimental unit with Equation $\boldsymbol{U}_2$
2. With the residuals grouped by period and the experimental unit, the $\boldsymbol{\Psi}$ matrix is estimated with $\hat{\psi}_{jj'}$, but discounting the correlation already captured by $\boldsymbol{R}(\boldsymbol{\alpha}_1)$.

Second result

Theorem 2

The estimator of $\boldsymbol{R}(\boldsymbol{\alpha})$ given by $\hat{\boldsymbol{\Psi}} \otimes \boldsymbol{R}_1(\hat{\boldsymbol{\alpha}}_1)$ is asymptotically unbiased and consistent.

Sketch of proof

1. Define the theoretical Pearson residuals of response of the kth of the ith experimental unit at the jth period as:

$$R_{ijk} = \frac{Y_{ijk} - \mu_{ijk}}{\phi\sqrt{V(\mu_{ijk})}}$$

2. Adapting the results of Cox and Snell (1968) and Cordeiro (2004) it is true that:

$$R_{ijk} \xrightarrow[n\to\infty]{d} N(0,1)$$

3. If the matrix $\boldsymbol{R} = (\boldsymbol{r}_1, \ldots, \boldsymbol{r}_n)_{P\times nL}$ with $\boldsymbol{r}_i$ of residuals of the ith individual is defined as:

$$\boldsymbol{r}_i = \begin{pmatrix} r_{i11} & r_{i21} & \cdots & r_{i1L} \\ r_{i21} & r_{i22} & \cdots & r_{i2L} \\ \vdots & \vdots & \ddots & \vdots \\ r_{iP1} & r_{i2P} & \cdots & r_{iPL} \end{pmatrix}$$

Sketch of proof

4. If the matrix $\boldsymbol{\Gamma}$ is defined as:

$$\boldsymbol{\Gamma}^T = \begin{pmatrix} \frac{1}{\sqrt{n}} & \vdots & \boldsymbol{G} \end{pmatrix}^T = \begin{pmatrix} \frac{1}{\sqrt{n}} & \frac{1}{\sqrt{n}} & \frac{1}{\sqrt{n}} & \frac{1}{\sqrt{n}} & \cdots & \frac{1}{\sqrt{n}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 & 0 & \cdots & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}} & 0 & \cdots & 0 \\ \frac{1}{\sqrt{12}} & \frac{1}{\sqrt{12}} & \frac{1}{\sqrt{12}} & -\frac{3}{\sqrt{12}} & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{1}{\sqrt{(n(n-1))}} & \frac{1}{\sqrt{n(n-1)}} & \frac{1}{\sqrt{n(n-1)}} & \frac{1}{\sqrt{n(n-1)}} & \cdots & -\frac{n-1}{\sqrt{n(n-1)}} \end{pmatrix}$$

5. Adapting the results of (Srivastava et al., 2008):

$$\boldsymbol{R}(\boldsymbol{\Gamma} \otimes \boldsymbol{I}_L) = \begin{pmatrix} \sqrt{n}\bar{\boldsymbol{r}} & \vdots & \boldsymbol{R}(\boldsymbol{G} \otimes \boldsymbol{I}_L) \end{pmatrix}$$

$$\boldsymbol{R}\left(\boldsymbol{I}_n \otimes \boldsymbol{\Psi}^{-1}\right)\boldsymbol{R}^T = n\bar{\boldsymbol{r}} \otimes \boldsymbol{\Psi}^{-1}\boldsymbol{R} + \boldsymbol{Z}(\boldsymbol{G}^T \otimes \boldsymbol{I}_L)\boldsymbol{Z}^T$$

Sketch of proof

6. By properties of R_{ijk} and the central limit theorem, we get:

$$E(Z_i) = \mathbf{0}_{P\times L} \text{ and } Var(Vec(Z_i)) = \Psi \otimes R_1(\alpha_1))$$

7.

$$E(z_{(j)1}, z_{(j)1}^T) = \Psi_{jj} R_1(\alpha_1) \qquad E(Z_1, Z_1^T) = (trace(\Psi)) R_1(\alpha_1)$$

8. Therefore, since $\Psi_{jj} = 1$, $\forall j = 1, \ldots, P$, $trace(\Psi) = 1$ we get:

$$E\left(R_1(\alpha_1)^{\frac{1}{2}} z_{(j')k} z_{(j)k}^T R_1(\alpha_1)^{\frac{1}{2}}\right) = \Psi_{jj'} I_P, \qquad \forall k = 1, \ldots, n-1$$

9. Futhermore,

$$Cov[R_1(\alpha_1)^{\frac{1}{2}} z_{(k)j} z_{(i)j}^T R_1(\alpha_1)^{\frac{1}{2}}, R_1(\alpha_1)^{\frac{1}{2}} z_{(k)j'} z_{(i)j'}^T R_1(\alpha_1)^{\frac{1}{2}}] = \mathbf{0}$$

□

Simulation study

Simulation

- Switch back crossover design: ABA, BAB
- $\boldsymbol{Y_{ijk} \sim Binomial(p_{ijk})}$; $\boldsymbol{i = 1, \ldots, n}$; $\boldsymbol{n \in \{5, 10, 100\}}$
- $\boldsymbol{ln\left(\frac{p_{ijk}}{1-p_{ijk}}\right) = \mu + \tau +}$ Period effect $+$ Carryover effect
- $\boldsymbol{\tau = \tau_B - \tau_A}$, and $\boldsymbol{\tau \in \{-2, -1.9, -1.8, \ldots, 1.8, 1.9, 2\}}$
- Period effect $\sim \boldsymbol{U(-1, 1)}$ and carry-over effect $\sim \boldsymbol{U(-1, 1)}$.
- $$\boldsymbol{\Psi} = \begin{pmatrix} 1 & \psi_{12} & \psi_{13} \\ \psi_{21} & 1 & \psi_{23} \\ \psi_{13} & \psi_{23} & 1 \end{pmatrix}, \qquad \psi_{ij} \sim \boldsymbol{U(-1, 1)} \text{ and } \boldsymbol{det(\Psi) > 0}$$
- $\boldsymbol{R_1(\alpha_1)}$, one of two structures was selected: exchangeable and autoregressive.
- Six models were fitted with the same linear predictor, but different correlation matrices

Simulation study

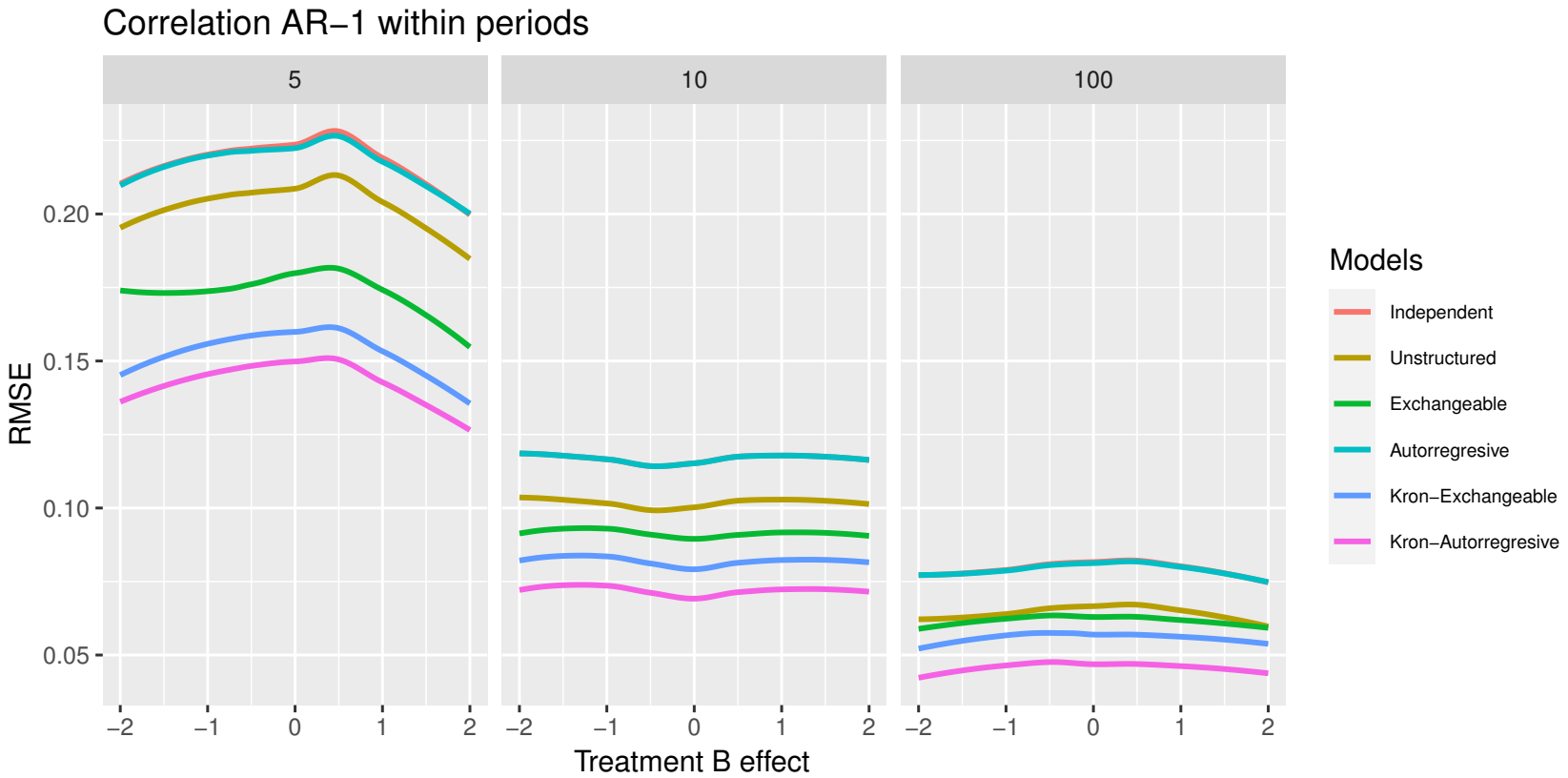


Figure: *RMSE for each value of the treatment effect. The average over all simulations where the correlation matrix was autoregressive and Kronecker product with* Ψ *is shown. Each subplot represents the number of individuals per sequence.*

Simulation study

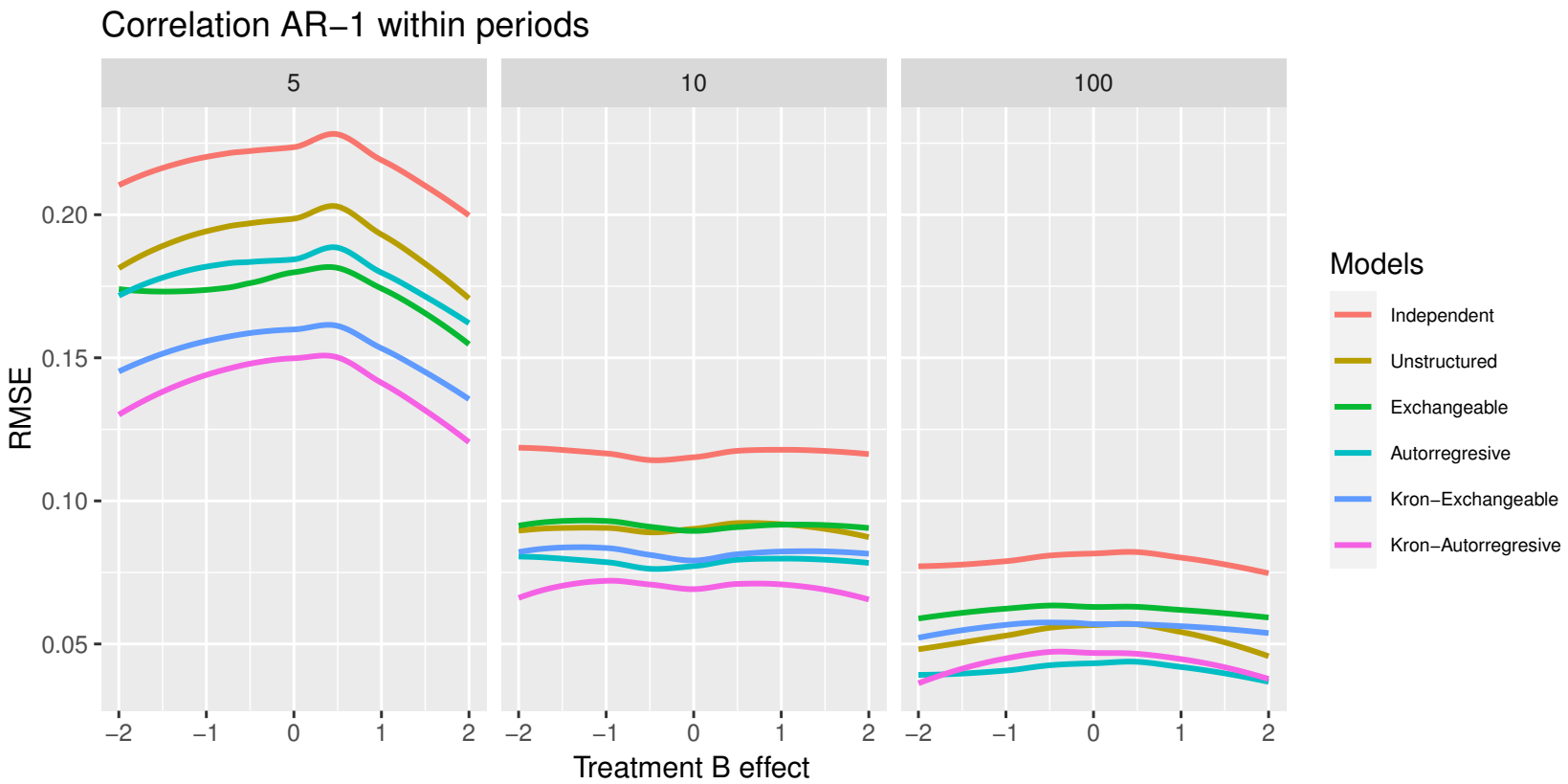


Figure: *RMSE for each value of the treatment effect. The average over all simulations where the correlation matrix was autoregressive and Kronecker product with* $\boldsymbol{\Psi} = \boldsymbol{I_3}$ *is shown. Each subplot represents the number of individuals per sequence.*

Correlation matrices for examples

Matriz $R(\alpha)$	*QIC* arterial	*QIC* cows
I_{PL}	46086	20.21
$I_P \otimes AR(1)_L$	44166	18.06
$I_P \otimes Exch_L$	45059	17.98
$AR(1)_{PL}$	44923	20.46
$Exch_{PL}$	44158	20.11
$\Psi \otimes AR(1)_L$	43400	14.68
$\Psi \otimes Exch_L$	43393	15.06

Table: *Correlation matrices and the corresponding* ***QIC*** *values*

$\Psi = I_P$

When it is assumed that $\Psi = I_P$, we are assuming that the observations of the same individual are independent between periods.

Correlation matrices

For the Dairy cattle

$$\hat{\boldsymbol{\Psi}} = \begin{pmatrix} 1.0000 & 0.3650 \\ 0.3650 & 1.0000 \end{pmatrix}, \hat{\boldsymbol{R}}_1(\boldsymbol{\alpha}) \begin{pmatrix} 1.0000 & 0.3362 & 0.3362^2 \\ 0.3362 & 1.0000 & 0.3362 \\ 0.3362^2 & 0.3362 & 1.0000 \end{pmatrix}$$

For the Arterial data

$$\hat{\boldsymbol{\Psi}} = \begin{pmatrix} 1.0000 & 0.0537 & 0.4486 \\ 0.0537 & 1.0000 & 0.3756 \\ 0.4486 & 0.3756 & 1.0000 \end{pmatrix}, \hat{\boldsymbol{R}}_1(\boldsymbol{\alpha}) = \begin{pmatrix} 1.0000 & 0.4958 & 0.4958 & \cdots & 0.4958 \\ 0.4958 & 1.0000 & 0.4958 & \cdots & 0.4958 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0.4958 & 0.4958 & 0.4958 & \cdots & 1.0000 \end{pmatrix}$$

Estimated parameters

For the Dairy cattle

Parameter	Estimate	Robust SE	Robust z	p-value
Intercept	1.9911	0.4579	18.9045	<0.001
Base	0.3707	0.1719	4.6515	0.0310
Time	-0.0086	0.0068	1.6093	0.2046
Period 2	0.2517	0.0565	19.8628	<0.001
Treatment A	-0.1438	0.0494	8.4638	0.0036
Carry-over A	-4.2480	1.4128	-3.0069	0.0026
Time^2	0.0001	0.0001	1.0223	0.3120

Table: *Estimates obtained by GEE for dairy cattle data*

Estimated parameters

For the Arterial data

Parameter	Estimate	Robust SE	Robust z	p-value
Intercept	105.1764	3.4255	30.7037	<0.001
Base	9.4286	2.2744	4.1455	<0.001
Time	0.0961	0.0364	2.6385	0.0042
Period 2	5.0793	3.5378	-1.4357	0.0755
Period 3	11.4009	7.4704	-1.5261	0.0635
Treatment B	-1.1014	1.8655	-0.5904	0.2775
Treatment C	-10.9991	3.2589	-3.3751	0.0004
Carry-over B	-4.5981	5.4023	0.8511	0.1973
Carry-over C	-11.5899	5.6351	2.0567	0.0199
Time^2	-0.0003	0.0001	-2.3884	0.0085

Table: *Estimates obtained by GEE for arterial pressure data*

The carryover effects are assumed constant over time

Semiparametric GEE

We propose a semiparametric model with B-splines for the time and carry-over effects as follows:

Correlation

$$E(Y_{ijk}) = \mu_{ijk}, \qquad Var(Y_{ijk}) = \phi V(\mu_{ijk})$$

$$g(\mu_{ijk}) = x_{ijk}^T \beta + f(Z_{ijk}) + \sum_{c=1}^{C} f_c(Z_{ijk})$$

$$V(\mu_i) = \left[D(V(\mu_{ijk})^{\frac{1}{2}}) R(\alpha) D(V(\mu_{ijk})^{\frac{1}{2}}) \right]_{L \times L}$$

where Z_{ijk} is the time at which the k-th observation of the i-th experimental unit was measured in the j-th period. f is a function describing the time's effect period, f_c is a function describing the previous treatment carry-over effect on the current period, with $f_c(Z_{i1k}) = 0$.

Carry-overs

For example in the blood pressure data set described in Table 2, $Z_{i11} = -30,\ Z_{i12} = -15, \ldots, Z_{i1\,10} = 240$, that is, the minutes of measurement within each period

Simple carry-over

$C = 2$, that is, f_1 is the carry-over effect of A, f_2 is the carry-over effect of B.

Complex carry-over

$C = 6$, that is, f_1 is the carry-over effect of A on B, f_2 is the carry-over effect of A on C, f_3 is the carry-over effect of B on A, f_4 is the carry-over effect of B on C, f_5 is the carry-over effect of C on A and f_6 is the carry-over effect of C on B

Advantages

The usual GEE modelling does not allow modelling complex carry-over effects, but this methodology does allow modelling them.

GEE splines

Let $\{s_1(t), \ldots, s_m(t)\}$ be a basis splines, then the f and f_c functions can be approximated through the following equations (Yu and Peace, 2012), respectively,

$$\hat{f}(t) = \sum_{b=1}^{m} \hat{\alpha}_{1b} s_b(t),\ \hat{f}_c(t) = \sum_{b=1}^{m} \hat{\alpha}_{cb} s_b(t), \forall c = 1, \ldots, C$$

the following generalized estimation equations are proposed for $\alpha_1 = \{\alpha_{11}, \ldots, \alpha_{1m}\}$, $\alpha_c = \{\alpha_{c1}, \ldots, \alpha_{cm}\}$, β y $R(\alpha)$:

For the time effect

$$U_1(\alpha_1, t|\beta, \alpha_2, \alpha) = \sum_{i=1}^{n} \left\{ diag\left(\frac{\partial \mu_{ijk}}{\partial \alpha_1}\right) \right\}_i \frac{V_{1i}^{-1}}{\phi} \left(y_i - \mu_i \left[X_i\beta, \sum_{b=1}^{m} \alpha_{1b} s_b(t), \hat{f}_c(Z_i) \right] \right)$$

GEE splines

For the carry-over effect

$$U_2(\boldsymbol{\alpha_c}, t|\boldsymbol{\beta}, \boldsymbol{\alpha_1}, \boldsymbol{\alpha}) = \sum_{i=1}^{n} \left\{ diag\left(\frac{\partial \mu_{ijk}}{\partial \boldsymbol{\alpha_2}}\right) \right\}_i \frac{\boldsymbol{V}_{2i}^{-1}}{\phi} \left(\boldsymbol{y}_i - \boldsymbol{\mu}_i \left(\boldsymbol{X}_i\boldsymbol{\beta}, \sum_{b=1}^{m} \alpha_{2b} s_b(t), \hat{f}(\boldsymbol{Z}_{1i}) \right) \right)$$

For $\boldsymbol{\beta}$ and correlation matrix

$$U_3(\boldsymbol{\beta}|\boldsymbol{\alpha_1}, \boldsymbol{\alpha_2}, \boldsymbol{\alpha}) = \sum_{i=1}^{n} \left\{ diag\left(\frac{\partial \mu_{ijk}}{\partial \boldsymbol{\beta}}\right) \right\}_i \frac{\boldsymbol{V}_{3i}^{-1}}{\phi} \left(\boldsymbol{y}_i - \boldsymbol{\mu}_i \left[\boldsymbol{X}_i\boldsymbol{\beta}, \hat{f}(\boldsymbol{Z}_i), \hat{f}_c(\boldsymbol{Z}_i) \right] \right)$$

$$U_4(\boldsymbol{\alpha}|\boldsymbol{\beta}, \boldsymbol{\alpha_1}, \boldsymbol{\alpha_2}) = \sum_{i=1}^{S} \sum_{k=1}^{n_i} \left(\frac{\partial \boldsymbol{\varepsilon}_{ik}}{\partial \boldsymbol{\alpha}} \right)^T \boldsymbol{F}_{ik}^{-1} \left(\boldsymbol{W}_{ik} - \boldsymbol{\varepsilon}_{ik} \right)$$

Steps of estimation extending the Fisher Scoring

1. Set initial values $\boldsymbol{\alpha}_2^{(0)}$, $\boldsymbol{\beta}^{(0)}$ and $\boldsymbol{\alpha}^{(0)}$
2. Find the value $\boldsymbol{\alpha}_1^{(1)}$ that solves the equation

$$U_1(\boldsymbol{\alpha_1}^{(1)}, t|\boldsymbol{\beta}^{(0)}, \boldsymbol{\alpha}_2^{(0)}, \boldsymbol{\alpha}^{(0)}) = 0$$

3. Find the value $\boldsymbol{\alpha}_2^{(1)}$ that solves the equation

$$U_2(\boldsymbol{\alpha_2}^{(1)}, t|\boldsymbol{\beta}^{(0)}, \boldsymbol{\alpha}_1^{(1)}, \boldsymbol{\alpha}^{(0)}) = 0$$

4. Find the value $\boldsymbol{\beta}^{(1)}$ that solves the equation

$$U_3(\boldsymbol{\beta}^{(1)}|\boldsymbol{\alpha}_1^{(1)}, \boldsymbol{\alpha}_2^{(1)}, \boldsymbol{\alpha}^{(0)}) = 0$$

5. Find the value $\boldsymbol{\alpha}^{(1)}$ that solves the equation

$$U_4(\boldsymbol{\alpha}^{(1)}|\boldsymbol{\beta}^{(1)}, \boldsymbol{\alpha}_1^{(1)}, \boldsymbol{\alpha}_2^{(1)}, \boldsymbol{\alpha}^{(0)}) = 0$$

6. Repeat steps (2) to (5) until convergence.

Estimate for $\boldsymbol{\beta}$

If solving the equation $\boldsymbol{U_3}$ with the Fisher scoring algorithm, in the $\boldsymbol{m}$-th step, the estimator of $\boldsymbol{\beta}$ is given by:

$$\boldsymbol{\beta}^{(m+1)} = \boldsymbol{\beta}^{(m)} - \left\{ \sum_{i=1}^{n} \boldsymbol{X}_i^T \boldsymbol{W}_i^{(m)} \boldsymbol{X}_i \right\}^{-1} \left\{ \sum_{i=1}^{n} \boldsymbol{X}_i^T \boldsymbol{W}_i^{(m)} \left(\boldsymbol{N}_i^{(m)} \right)^{-1} \boldsymbol{u}_i^{(m)} \right\}$$

$\hat{\boldsymbol{\beta}}$ is like a weighted least squares estimator

$$\boldsymbol{\beta}^{(m+1)} = \left\{ \sum_{i=1}^{n} \boldsymbol{X}_i^T \boldsymbol{W}_i^{(m)} \boldsymbol{X}_i \right\}^{-1} \left\{ \sum_{i=1}^{n} \boldsymbol{X}_i^T \boldsymbol{W}_i^{(m)} \boldsymbol{z}_i^{(m)} \right\}$$

where

$$\boldsymbol{z}_i^{(m)} = \boldsymbol{X}_i \boldsymbol{\beta}^{(m)} - \boldsymbol{N}_i^{-1(m)} \boldsymbol{u}_i^{(m)}$$

Estimate for $\boldsymbol{\beta}$

where

$$\boldsymbol{W}_i^{(m)} = \left\{ diag\left(\frac{\partial \mu_{ijk}^{(m)}}{\partial \boldsymbol{\beta}^{(m)}} \right) \right\}_i \boldsymbol{V}_{3i}^{-1} \left\{ diag\left(\frac{\partial \mu_{ijk}^{(m)}}{\partial \boldsymbol{\beta}^{(m)}} \right) \right\}_i^{-1}$$

$$\boldsymbol{u}_i^{(m)} = \left(\boldsymbol{y}_i - \boldsymbol{\mu}_i^{(m)} \left[\boldsymbol{X}_i \boldsymbol{\beta}^{(m)}, \hat{f}(\boldsymbol{Z}_i^{(m)}, \hat{f}_c(\boldsymbol{Z}_i^{(m)}) \right] \right)$$

$$\boldsymbol{N}_i^{(m)} = \left\{ diag\left(\frac{\partial \mu_{ijk}^{(m)}}{\partial \boldsymbol{\beta}^{(m)}} \right) \right\}_i$$

$$\boldsymbol{A} = \sum_{i=1}^{n} \boldsymbol{N}_i \boldsymbol{V}_{3i}^{-1} \boldsymbol{N}_i^T$$

$$\boldsymbol{B} = \sum_{i=1}^{n} \boldsymbol{N}_i \boldsymbol{V}_{3i}^{-1} \left(\boldsymbol{y}_i - \hat{\boldsymbol{\mu}}_i \right) \left(\boldsymbol{y}_i - \hat{\boldsymbol{\mu}}_i \right)^T \boldsymbol{V}_{3i}^{-1} \boldsymbol{N}_i^T$$

Third result

Theorem 3

Under the assumption that the r-th derivative of f and f_c is bounded for some $r \geq 2$ and that the number of knots $m = m_n \to \infty$, but $\frac{m}{n} \to 0$ then $\hat{\beta} - \beta \xrightarrow{n \to \infty} 0$. Also, if $m = O\left(n^{\frac{1}{(2r+1)}}\right)$ then:

$$\frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{n_i} \left\{ \sum_{b=1}^{m} \hat{\alpha}_{1b} s_b(Z_{ijk}) - f(Z_{ijk}) \right\}^2 = O\left(n^{-\frac{2r}{(2r+1)}}\right)$$

$$\frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{n_i} \left\{ \sum_{b=1}^{m} \hat{\alpha}_{cb} s_b(Z_{ijk}) - f_c(Z_{ijk}) \right\}^2 = O\left(n^{-\frac{2r}{(2r+1)}}\right)$$

$$\sqrt{n}(\hat{\beta} - \beta) \to N(0, A^{-1} B A^{-1})$$

Sketch of proof

1. We assume that there exists a random variables, δ_{ijk}, with $E(\delta_{ijk}) = 0$ and $Var(\delta_{ijk}) \leq \infty$ and continuous functions $g_1, \ldots, g_m$ such that:

$$x_{ijkl} = g_l(Z_{ijk}) + \delta_{ijkl}$$

2. Since Y_{ijk} belongs to the exponential family and by Liang and Zeger (1986), and Lehmann and Casella (2006), then:

$$E(U_1(\epsilon_i, t)) = 0 \, \forall t \in \mathbb{R}$$
$$0 < E\left(U_1(\epsilon_i, t) U_1(\epsilon_i, t)^T\right) < \infty$$
$$E(U_2(\epsilon_i, t)) = 0, \qquad 0 < E\left(U_2(\epsilon_i, t) U_2(\epsilon_i, t)^T\right) < \infty, \ \forall t \in \mathbb{R}$$
$$E(U_3(\epsilon_i, t)) = 0, \qquad 0 < E\left(U_3(\epsilon_i, t) U_3(\epsilon_i, t)^T\right) < \infty, \ \forall t \in \mathbb{R}$$
$$E(U_4(\epsilon_i)) = 0, \qquad 0 < E\left(U_4(\epsilon_i) U_4(\epsilon_i)^T\right) < \infty$$

Sketch of proof

3. when $\boldsymbol{s \to 0}$, the following properties hold, respectively:

$$\sup_{ijk} |E(U_1(\epsilon_{ijk} + s, t)) - b_{1ijk}s| = O(s^2), \ \forall t \in \mathbb{R},$$

$$\sup_{ijk} |E(U_2(\epsilon_{ijk} + s, t)) - b_{2ijk}s| = O(s^2), \ \forall t \in \mathbb{R},$$

$$\sup_{ijk} |E(U_3(\epsilon_{ijk} + s)) - b_{3ijk}s| = O(s^2),$$

4. When $\boldsymbol{s \to 0}$, there exist constants $\boldsymbol{c > 0}$ and, $\boldsymbol{C < \infty}$ such that:

$$\sup_{ijk} \left\{ E(U_1(\epsilon_{ijk} + s, t) - U_1(\epsilon_{ijk}, t))^2 \right\} \leq C|s|, \ \forall t \in \mathbb{R}$$

$$\sup_{ijk} \left\{ E(U_2(\epsilon_{ijk} + s, t) - U_2(\epsilon_{ijk}, t))^2 \right\} \leq C|s|, \ \forall t \in \mathbb{R}$$

$$\sup_{ijk} \left\{ E(U_3(\epsilon_{ijk} + s) - U_1(\epsilon_{ijk}))^2 \right\} \leq C|s|$$

4. And using Theorem 1 and Theorem 2 of He et al. (2002), the theorem are obtained. □

Simulation study

Simulation

- Switch back crossover design: ABA, BAB
- $Y_{ijk} \sim Poisson(\mu_{ijk})$; $i = 1, \ldots, n$; $n \in \{3, 5, 10, 15, \ldots, 100\}$
- $ln(\mu_{ijk}) = \mu + \tau +$ Period effect $+$ Time effect $+$ Carryover effect
- $\tau = \tau_B - \tau_A$, and $\tau \in \{-2, -1.9, -1.8, \ldots, 1.8, 1.9, 2\}$
- Period effect $= 3$, and $L = 15$
- Time effect $= c_1 cos(t_{jk})$
- Carryover effect $= c_2 sen(t_{jk})\delta_{jk}$
- $c_1 \sim N(0, 1)$, $c_2 \sim N(0, 1)$
- Three models were adjusted for each scenario:
  1. A model with semiparametric GEE denoted by GEE-S
  2. A GEE model where the effect of time is linear which is denoted by GEE-1
  3. A GEE model with quadratic time effect which is denoted by GEE-2

Simulation study

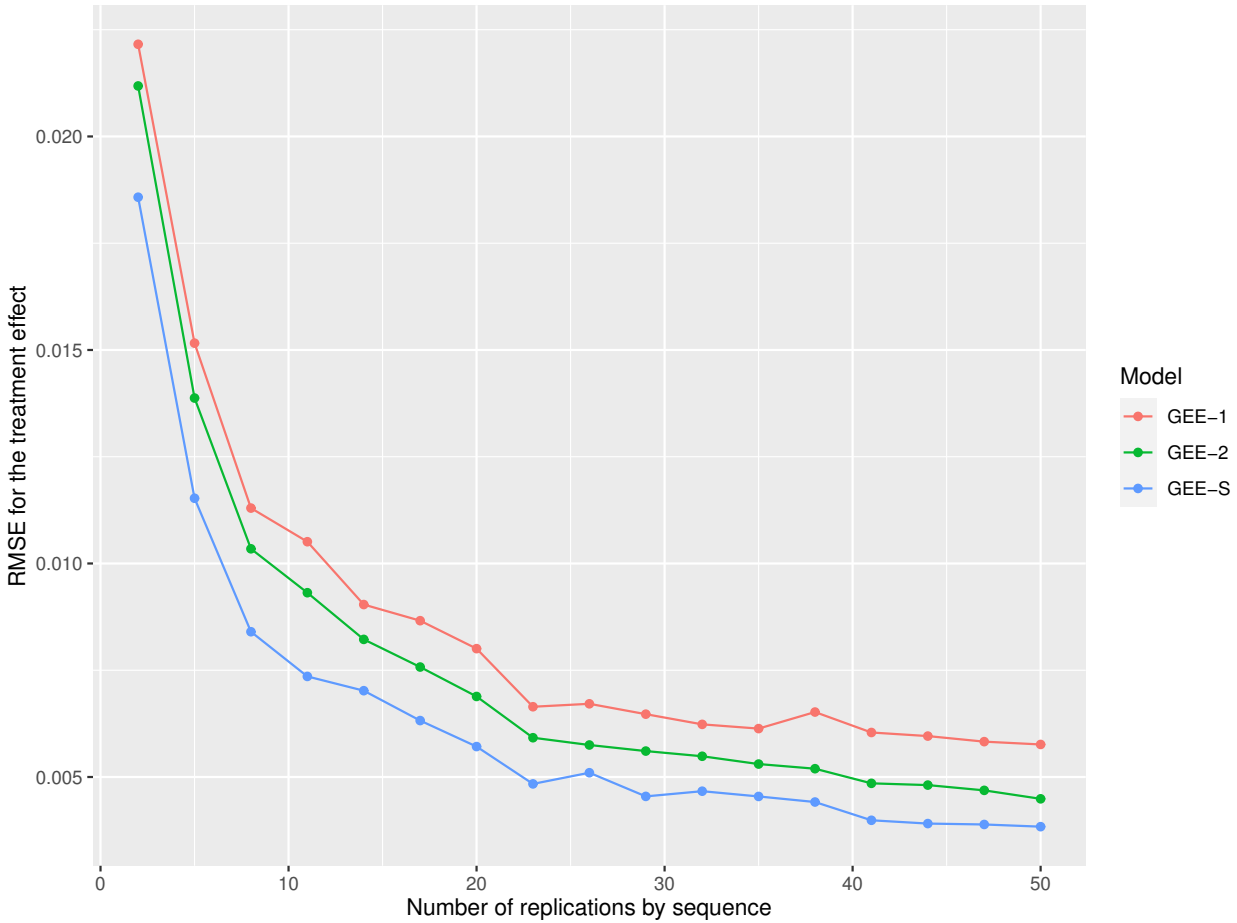


Figure: *RMSE for the estimate of* $\beta_1 = 0.5$ *in each of the models*

Arterial data

Matriz $R(\alpha)$	*QIC* GEE	*QIC* Semiparametric GEE
I_{PL}	46086	42646
$AR(1)_{PL}$	44923	42364
$Exch_{PL}$	44158	43213
R_{PL}	100107	47246

Table: *Correlation matrices and the corresponding* ***QIC*** *values*

Dairy cattle

In the case of cow data, with only 3 observations per period, it is impossible to fit the non-parametric model. In simulation exercises, it was found that at least 5 observations are needed per period.

Semiparametric effects (Simple carry-over)

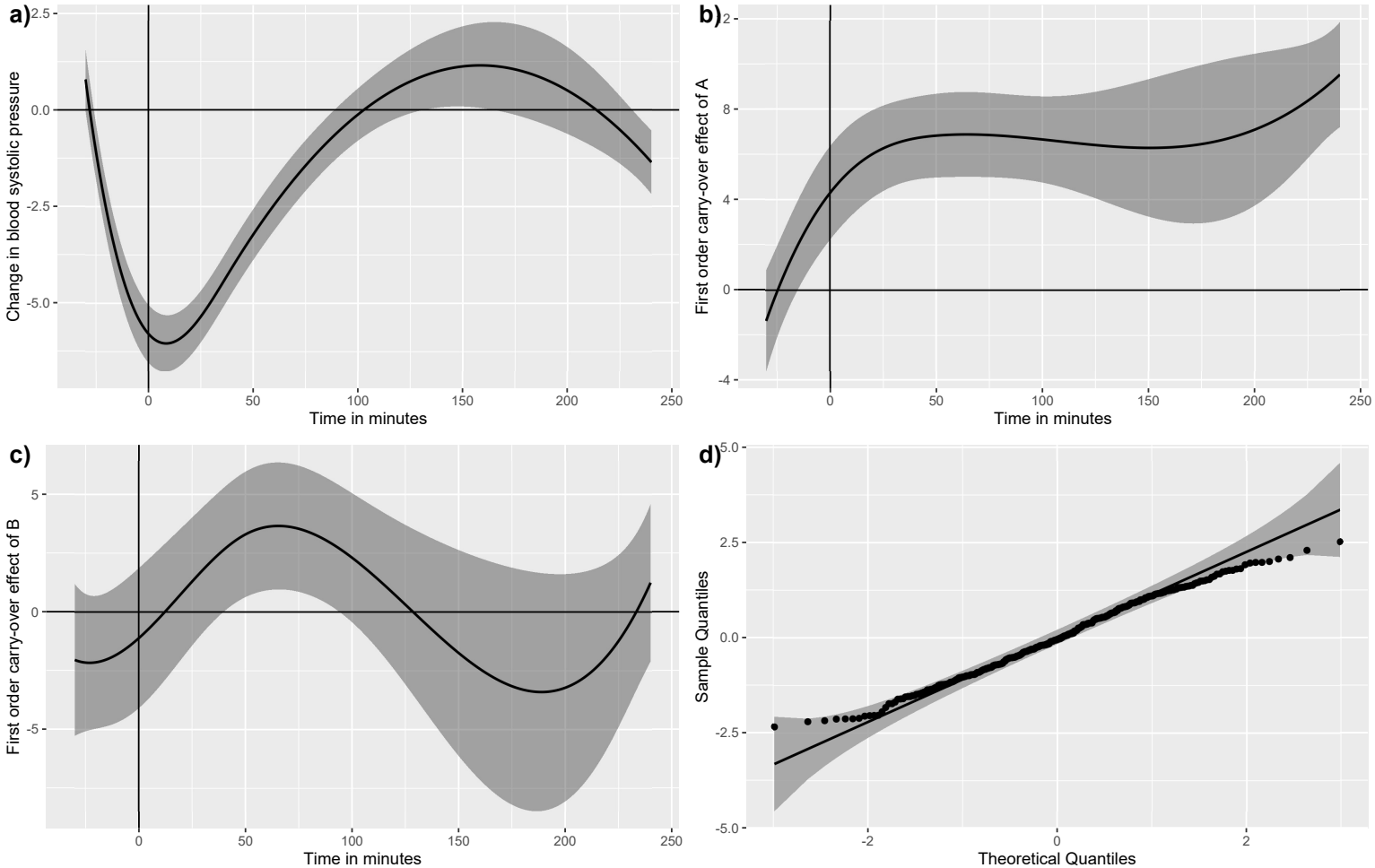


Figure: *a) Changes of blood systolic pressure, b) First order carry-over effect of the treatment A, c) First order carry-over effect of the treatment B, and d) Residuals normal probability plot. All figures present 95% confidence intervals*

Semiparametric effects (Simplex carry-over)

Parametric estimation

Parameter	Estimate	Std.err	Wald	Pr(> |$\boldsymbol{W}$|)
Intercept	107.2944	3.3474	1027.4000	0.0000
Base	3.8505	1.3484	8.1542	0.0043
Period 2	-1.0462	3.4939	0.0897	0.7646
Period 3	-1.8525	3.6337	0.2599	0.6102
Treatment B	3.6068	2.9630	1.4818	0.2235
Treatment C	-5.8752	2.6404	1.1331	0.0261

Table: *Analysis of blood systolic pressure data using GEE-Splines*

Conclusions

1. Using GEE, unbiased estimates of important design effects are obtained, regardless of the presence or absence of carryover effects, in addition to obtaining a deviance analysis table for design analysis.
2. The family of correlation structures built using the Kronecker product allows greater flexibility in the correlation structure and helps to better analyze crossover designs.
3. The semiparametric methodology provides an estimate of the effects over time, and reduces the bias of the resulting estimators. It also allows making asymptotic inferences and better modelling temporary carryover behaviors that would be intractable in parametric scenarios.

Future Research

1. Check the theoretical and practical performance of the combination of the Kronecker structure with the semiparametric modelling (In progress).
2. Build a semi-parametric model that allows you to estimate complex carryover effects (In progress).
3. Propose a methodology to deal with missing data, either due to random missing data or non-ignorable dropout.
4. Implement a Bayesian approach to work out carryover effects and possible treatment interactions over time.

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Accepted and Published Papers

Papers

1. Deviance Analysis: (Cruz et al., 2023a) Cruz Gutierrez NA, Melo OO & Lopez LA. Analysis of crossover experiments with count data in the presence of carry-over effects. *Statistica Neerlandica*. 2023;0(0). `https://doi:10.1111/stan.12295`
2. Kronecker Correlation: (Cruz et al., 2023b) Cruz, N.A., Melo, O.O. & Martinez, C.A. A correlation structure for the analysis of Gaussian and non-Gaussian responses in crossover experimental designs with repeated measures. *Statistical Papers* (2023) 32(5), 1033-1050. `https://doi.org/10.1007/s00362-022-01391-z`
3. Semiparametric GEE: (Cruz et al., 2023c) Cruz Gutierrez NA, Melo OO & Martinez CA. Semiparametric generalized estimating equations for repeated measurements in crossover designs. *Statistical Methods in Medical Research*. 2023;0(0). `https://doi:10.1177/09622802231158736`

R package

CrossCarry: Analysis of Data from a Crossover Design with GEE

Analyze data from a crossover design using generalized estimation equations (GEE), including carryover effects and various correlation structures based on the Kronecker product. It contains functions for semiparametric estimates of carry-over effects in repeated measures and allows estimation of complex carry-over effects. Related work includes: a) Cruz N.A., Melo O.O., Martinez C.A. (2023). "CrossCarry: An R package for the analysis of data from a crossover design with GEE". <arXiv:2304.02440v1>. b) Cruz N.A., Melo O.O., Martinez C.A. (2023). "A correlation structure for the analysis of Gaussian and non-Gaussian responses in crossover experimental designs with repeated measures". <doi:10.1007/s00362-022-01391-z> and c) Cruz N.A., Melo O.O., Martinez C.A. (2023). "Semiparametric generalized estimating equations for repeated measurements in cross-over designs". <doi:10.1177/09622802231158736>.

Version:	0.1.0
Depends:	R (≥ 4.0)
Imports:	dplyr, gee, geepack, ggplot2, splines, stats
Suggests:	testthat (≥ 3.0.0)
Published:	2023-04-11
Author:	Nelson Alirio Cruz Gutierrez [aut, cre, cph], Oscar Orlando Melo [aut], Carlos Alberto Martinez [aut]
Maintainer:	Nelson Alirio Cruz Gutierrez <neacruzgu at unal.edu.co>
License:	GPL (≥ 3)
NeedsCompilation:	no
Citation:	CrossCarry citation info
CRAN checks:	CrossCarry results

Figure: *https://cran.r-project.org/web/packages/CrossCarry/index.html, published in 11-04-2023*

Events

1. **31 SIMPOSIO INTERNACIONAL DE ESTADÍSTICA**: Modelamiento semiparametrico de efectos carryover en diseños crossover con medidas repetidas, Manizales, Colombia, 29 de Agosto a 2 de Septiembre de 2022.
2. **JORNADA AVANCES EN MÉTODOS ESTADISTICOS**: Una estructura de correlación de Kronecker para el análisis de respuestas en diseños experimentales crossover con GEE, 16 y 17 de febrero de 2023.
3. **XIX Conferencia Española de Biometría (CEB) y VIII Encuentro Iberoamericano de Biometría (EIB)**: Semi-parametric generalized estimating equations for repeated measurements in crossover designs, 27 al 30 de Junio de 2023, Vigo, Galicia, España.

The end

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Nelson Alirio Cruz